

Chapter 7. Limits and Continuity in Several Variables

Path dependence, continuity, and local stability

In single-variable calculus, the limit

$$\lim_{x \rightarrow a} f(x)$$

asks what happens as x approaches a from the left and from the right. In multivariable calculus, the limit

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y)$$

asks what happens as the point (x,y) approaches (a,b) from every possible direction and along every possible path. This is the first major conceptual difference between single-variable and multivariable calculus.

This chapter is about learning how to decide when a multivariable limit exists, when it fails to exist, and why continuity is the local stability condition that makes differentiation and integration behave well.

Connection

This chapter develops limits and continuity of functions of several variables, including the distance-based definition, path tests, examples such as $\frac{x^2-y^2}{x^2+y^2}$ and $\frac{xy^2}{x^2+y^4}$, and continuous examples such as $e^{-xy} \sin(x+y)$ and $\frac{\sin(x^2+y^2)}{x^2+y^2}$.

Learning goals

After studying this chapter, you should be able to:

1. explain the meaning of $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$;
2. use direct substitution when continuity applies;
3. show that a limit does not exist by finding two paths with different limiting values;
4. understand why checking only lines is sometimes not enough;
5. prove simple limits using estimates, polar coordinates, and the squeeze theorem;
6. define continuity for functions of several variables;
7. identify common sources of discontinuity;
8. use numerical and graphical evidence responsibly;
9. connect continuity to stability in statistics, data science, and machine learning.

1 Why multivariable limits are different

A single-variable limit has only two basic directions of approach:

$$x \rightarrow a^- \quad \text{and} \quad x \rightarrow a^+.$$

A two-variable limit has infinitely many possible approaches:

$$(x, y) \rightarrow (a, b).$$

The point (x, y) can approach (a, b) along a horizontal line, a vertical line, a slanted line, a parabola, a spiral, or a more complicated curve. For the limit to exist, *all paths must lead to the same value*.

Core idea

To prove a multivariable limit exists, we must control all possible approaches. To prove a multivariable limit does not exist, it is enough to find two approaches that give different answers.

A useful analogy is hiking toward the summit of a mountain. If all trails leading to the summit approach the same height, the height has a well-defined limit. If different trails approach different heights near the same map location, then the surface does not have a well-defined limiting height there.

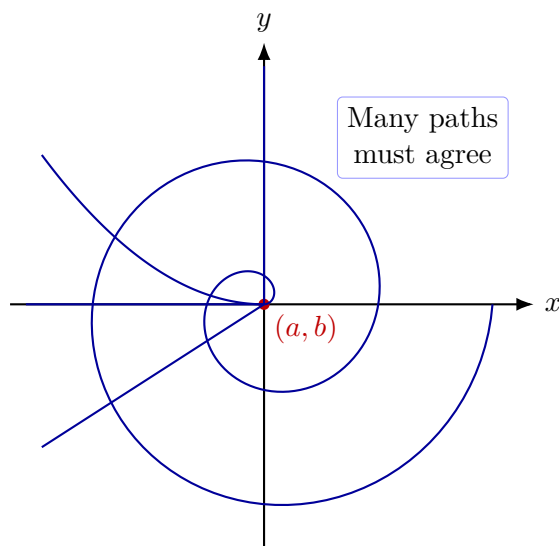


Figure 1: In two variables, a point can approach the target along infinitely many paths.

2 The formal definition

Let f be a function of two variables whose domain contains points arbitrarily close to (a, b) . We write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

if the values $f(x, y)$ become arbitrarily close to L whenever (x, y) is sufficiently close to (a, b) , but not equal to (a, b) .

Using distance, this means:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

Here ε measures closeness in output, and δ measures closeness in input.

Practical interpretation

The formal definition says that the output can be forced close to L by taking the input point close enough to (a, b) , regardless of the direction of approach.

The same definition works in $\mathbb{R}^n$. If $x = (x_1, \dots, x_n)$ and $a = (a_1, \dots, a_n)$, then

$$\|x - a\| = \sqrt{(x_1 - a_1)^2 + \dots + (x_n - a_n)^2}.$$

Then

$$\lim_{x \rightarrow a} f(x) = L$$

means that $f(x)$ is close to L whenever $\|x - a\|$ is small and nonzero.

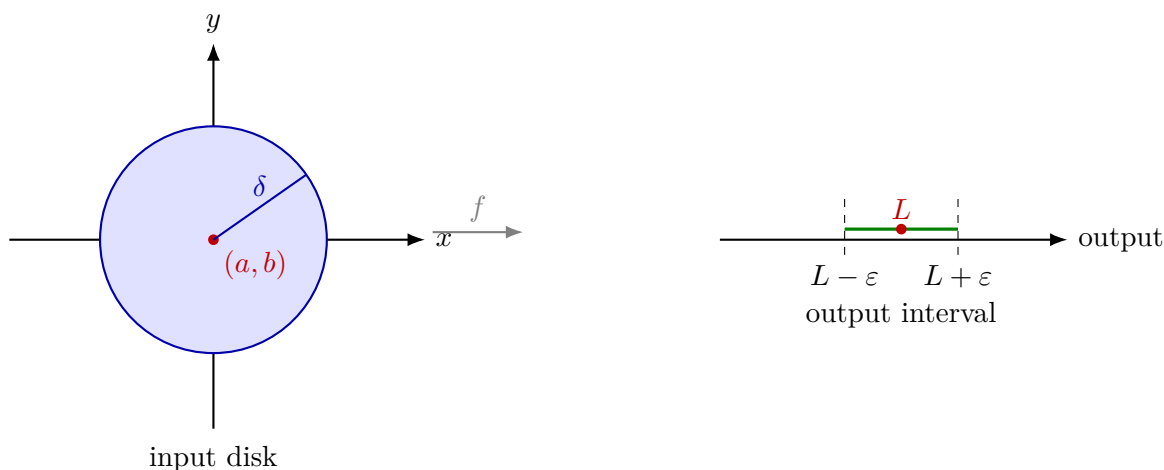


Figure 2: The ε - δ definition controls all points in a small punctured disk around the input point.

3 Easy limits by continuity

Many limits are easy because the function is continuous. Polynomials, exponential functions, trigonometric functions, and compositions of continuous functions are continuous wherever they are defined.

Example 3.1: Direct substitution

Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} e^{-xy} \sin(x+y).$$

Both e^{-xy} and $\sin(x+y)$ are continuous, so their product is continuous. Therefore the limit is obtained by substitution:

$$\lim_{(x,y) \rightarrow (0,0)} e^{-xy} \sin(x+y) = e^0 \sin(0) = 0.$$

Example 3.2: A rational function at a regular point

Evaluate

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x + y^2}{x^2 - y^2}.$$

The denominator at $(1, 2)$ is

$$1^2 - 2^2 = -3 \neq 0.$$

So the function is continuous at $(1, 2)$, and

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x + y^2}{x^2 - y^2} = \frac{1 + 2^2}{1^2 - 2^2} = -\frac{5}{3}.$$

When substitution works

If a formula is built from continuous operations and is defined at the target point, substitution is usually the first method to try.

4 Path tests for nonexistence

To show a limit does not exist, find two paths approaching the same point that give different limiting values.

Example 4.1: A classic path-dependent limit

Consider

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Along the path $y = 0$,

$$f(x, 0) = \frac{x^2}{x^2} = 1,$$

so $\lim_{x \rightarrow 0} f(x, 0) = 1$. Along the path $x = 0$,

$$f(0, y) = \frac{-y^2}{y^2} = -1,$$

so $\lim_{y \rightarrow 0} f(0, y) = -1$. The two paths give different values. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

does not exist.

Along a general line $y = mx$ with $x \neq 0$,

$$f(x, mx) = \frac{x^2 - m^2x^2}{x^2 + m^2x^2} = \frac{1 - m^2}{1 + m^2}.$$

The result depends on m , the slope of the path. That is a strong sign that the full two-variable limit does not exist.

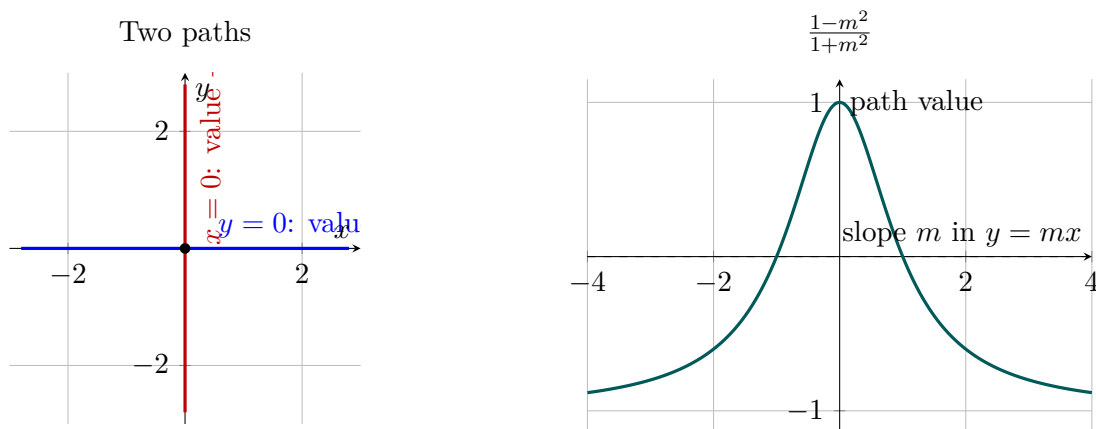


Figure 3: Different straight-line paths give different limiting values for $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$.

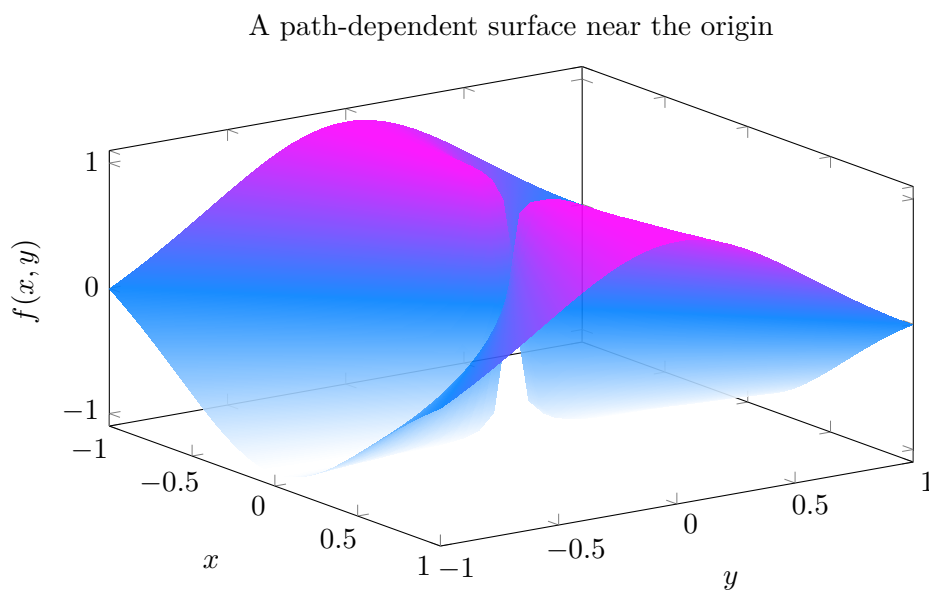


Figure 4: Near the origin, the surface approaches different heights from different directions.

5 Checking only lines is not enough

Sometimes every straight-line path gives the same value, but the limit still does not exist. This is one of the most important warnings in multivariable calculus.

Example 5.1: Lines are misleading

Consider

$$g(x, y) = \frac{xy^2}{x^2 + y^4}.$$

Along a straight line $y = mx$,

$$g(x, mx) = \frac{x(m^2x^2)}{x^2 + m^4x^4} = \frac{m^2x}{1 + m^4x^2} \rightarrow 0.$$

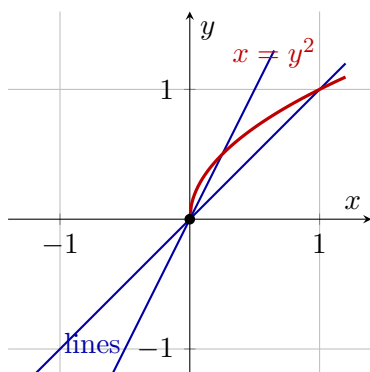
So every line $y = mx$ gives limiting value 0.

But along the curved path $x = y^2$,

$$g(y^2, y) = \frac{y^2y^2}{(y^2)^2 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}.$$

Therefore the full limit does not exist.

Straight and curved paths



Path values

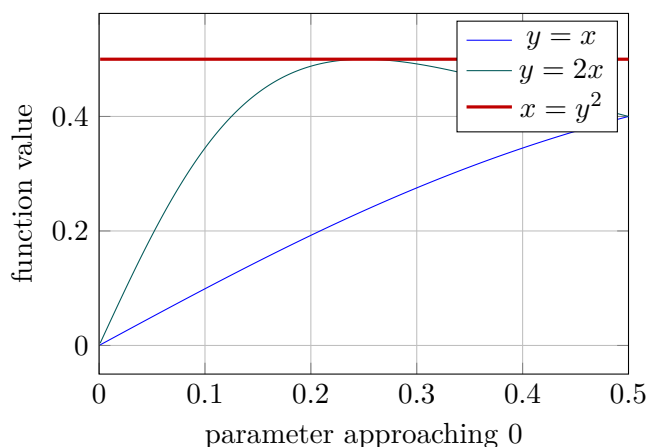


Figure 5: Line paths can all approach zero while a curved path approaches a different value.

Common mistake

Showing that several paths give the same value does not prove a limit exists. It only provides evidence. To prove existence, we need an argument that controls all paths.

6 Proving limits with estimates

To prove that a limit exists, we often use inequalities. The idea is to bound the complicated expression by something simpler that goes to zero.

A common strategy near $(0,0)$ is to use

$$r = \sqrt{x^2 + y^2}.$$

Then

$$|x| \leq r, \quad |y| \leq r.$$

If we can show

$$|f(x, y) - L| \leq Cr^p$$

for constants $C > 0$ and $p > 0$, then $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (0, 0)$.

Example 6.1: A squeeze theorem proof

Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

For $(x, y) \neq (0, 0)$,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{|x|^2 |y|}{x^2 + y^2}.$$

Since $x^2 \leq x^2 + y^2$, we have

$$\frac{|x|^2}{x^2 + y^2} \leq 1.$$

Therefore

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y|.$$

As $(x, y) \rightarrow (0, 0)$, we have $|y| \rightarrow 0$. By the squeeze theorem,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

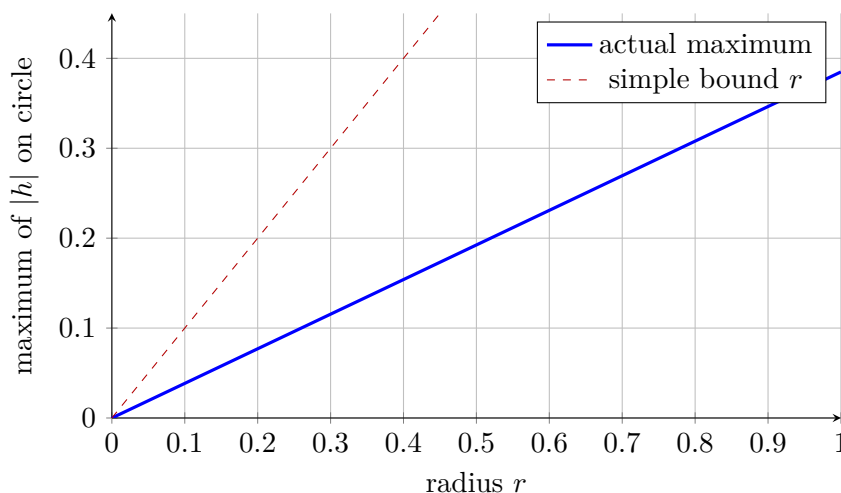


Figure 6: For $h(x, y) = \frac{x^2 y}{x^2 + y^2}$, the maximum absolute value on circles of radius r decreases linearly toward zero.

7 Polar coordinates as a limit tool

For limits at the origin in the plane, polar coordinates are often useful:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then $(x, y) \rightarrow (0, 0)$ means $r \rightarrow 0$. The angle θ may change in any way. If the expression becomes something that goes to L independently of θ , then the limit is L .

Example 7.1: A polar-coordinate proof

Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0.$$

Using $x = r \cos \theta$ and $y = r \sin \theta$,

$$\frac{xy}{\sqrt{x^2 + y^2}} = \frac{r^2 \cos \theta \sin \theta}{r} = r \cos \theta \sin \theta.$$

Since $|\cos \theta \sin \theta| \leq 1$,

$$|r \cos \theta \sin \theta| \leq r.$$

As $r \rightarrow 0$, this tends to 0. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0.$$

A limit controlled by distance to the origin

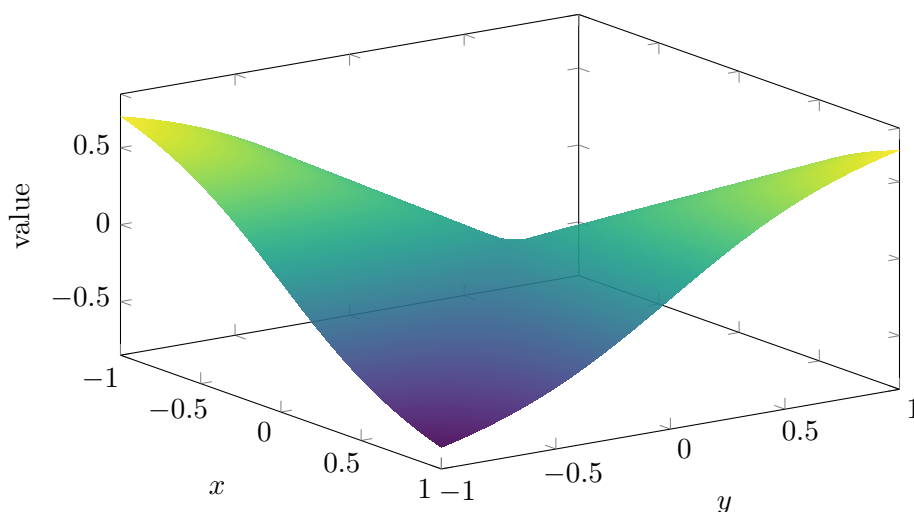


Figure 7: The polar form $r \cos \theta \sin \theta$ is controlled by the radius r .

8 A removable discontinuity

Consider

$$q(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

The expression is undefined at $(0, 0)$, but the limit exists. Let

$$u = x^2 + y^2.$$

As $(x, y) \rightarrow (0, 0)$, we have $u \rightarrow 0^+$. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{u \rightarrow 0^+} \frac{\sin u}{u} = 1.$$

The function is not continuous at the origin because it is not defined there. But we can make it continuous by defining

$$q(0, 0) = 1.$$

This is called a *removable discontinuity*.

A removable discontinuity filled in continuously

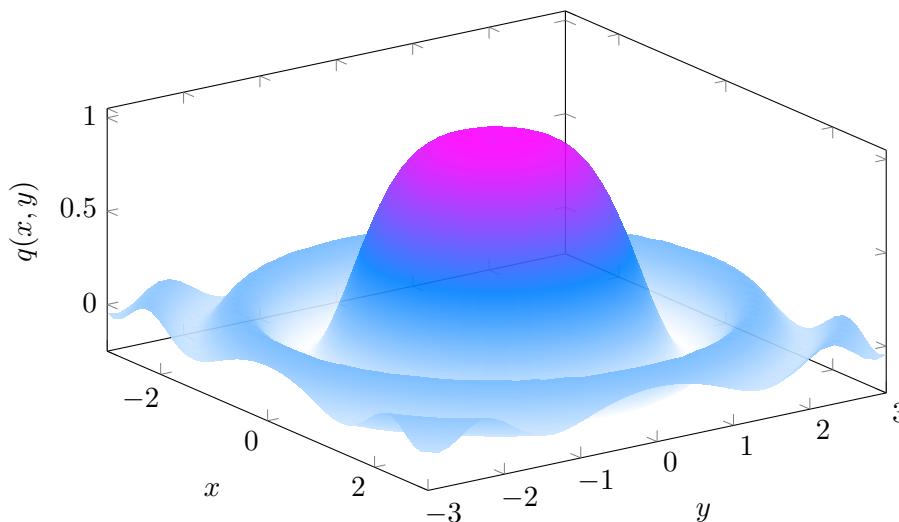


Figure 8: The function $\frac{\sin(x^2+y^2)}{x^2+y^2}$ has limiting value 1 at the origin.

9 Continuity in several variables

A function $f(x, y)$ is *continuous at* (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

This requires three things:

1. $f(a, b)$ is defined;
2. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists;
3. the limit equals the function value.

A function is continuous on a region D if it is continuous at every point of D .

Common sources of discontinuity

Source	Example	What can go wrong?
Denominator zero	$\frac{1}{x^2 + y^2}$	infinite blow-up near the origin
Path dependence	$\frac{x^2 - y^2}{x^2 + y^2}$	different paths give different values
Removable hole	$\frac{\sin(x^2 + y^2)}{x^2 + y^2}$	formula undefined at one point, but limit exists
Domain boundary	$\sqrt{1 - x^2 - y^2}$	function exists only on a disk
Piecewise rule	threshold or classifier	jump discontinuity across a boundary

Continuity and modeling

A continuous model changes outputs gradually when inputs are changed slightly. This is essential for numerical stability, optimization, interpolation, and sensitivity analysis.

10 Continuity on domains and boundaries

Some functions are continuous on their natural domains even though the domain has a boundary. For example,

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

is defined on the disk

$$D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

It is continuous at every point of this disk relative to its domain. Near boundary points, we only approach through points in the domain.

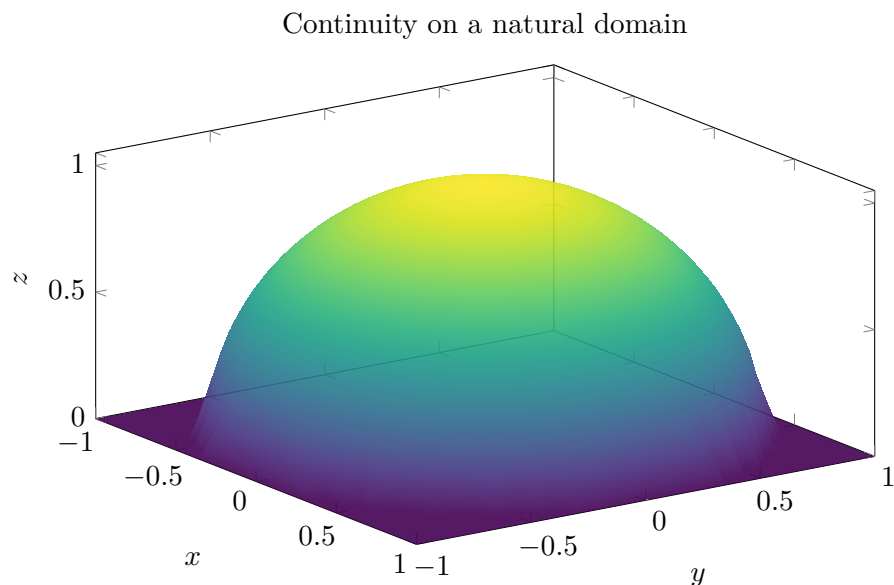


Figure 9: The upper hemisphere is continuous on the closed disk $x^2 + y^2 \leq 1$.

This distinction matters in applications. A probability density may be defined only on a simplex, a loss function may be meaningful only on feasible parameters, and a physical model may be valid only inside a spatial region.

11 Numerical experiments with limits

Numerical exploration can suggest whether a limit exists, but it cannot by itself prove a limit. A computer can sample many paths, but it cannot sample all paths.

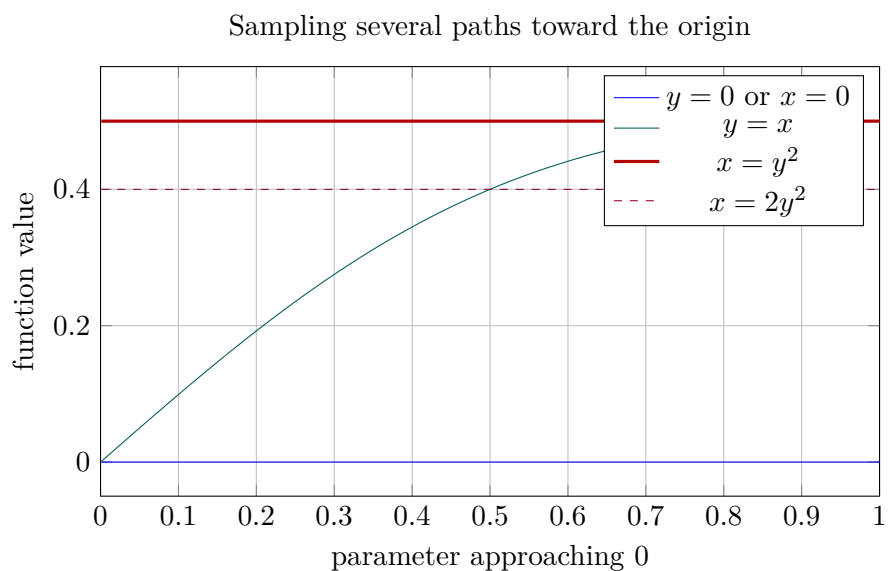


Figure 10: Numerical path sampling is useful evidence but not a proof.

Computational caution

A graph can miss a narrow ridge, a sharp spike, or a curved path where behavior changes. Use computation to develop intuition, then use algebra to justify the conclusion.

12 Higher-dimensional continuity

For a function

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

continuity at $a \in \mathbb{R}^n$ means

$$\lim_{x \rightarrow a} f(x) = f(a).$$

The same distance idea applies:

$$\|x - a\| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

For example, the squared norm function

$$f(x) = \|x\|^2 = x_1^2 + \cdots + x_n^2$$

is continuous on $\mathbb{R}^n$. The logistic regression loss, least-squares loss, and many smooth neural-network losses are also continuous functions of their parameters.

Example 12.1: Least-squares loss is continuous

Let A be a fixed data matrix and b a fixed response vector. Define

$$L(w) = \|Aw - b\|^2.$$

This is a polynomial function of the components of w , so it is continuous. Small changes in w cause small changes in $L(w)$. This is one reason gradient-based methods can be used to minimize least-squares loss.

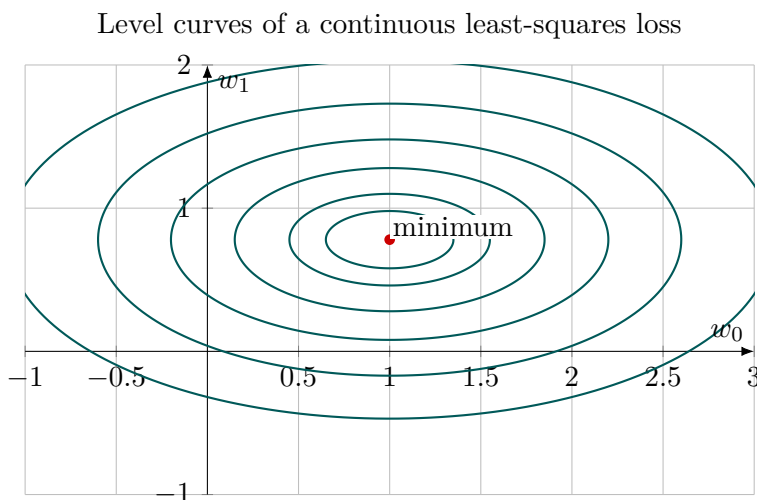


Figure 11: A least-squares loss surface is continuous and has smooth level curves.

13 Modern applications

Statistics

In statistics, likelihood and loss functions are often functions of parameters. Continuity allows us to use local approximations, optimization methods, and asymptotic arguments. For example, if a likelihood changes continuously with parameters, then nearby parameter values produce nearby likelihood values.

Data science

A data record may be a vector

$$x = (x_1, \dots, x_n).$$

A prediction model may be a function

$$f(x_1, \dots, x_n).$$

Continuity means that small changes in the input features produce small changes in the prediction. This is desirable for many regression models.

Machine learning and AI

Training a model usually means minimizing a loss function

$$L(\theta).$$

Continuity of L is the first requirement for meaningful optimization. Differentiability, which comes next, is stronger. A function can be continuous without being differentiable, but it cannot be differentiable without being continuous.

Bridge to gradients

The next chapters build on continuity. Partial derivatives, tangent planes, gradients, and optimization all require careful local behavior near a point.

14 Worked examples summary

Example type	Limit	Reason
Exists by continuity	$\lim_{(x,y) \rightarrow (1,2)} \frac{x+y^2}{x^2-y^2} = -\frac{5}{3}$	denominator is nonzero at $(1, 2)$
Does not exist by two paths	$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2-y^2}{x^2+y^2}$	$y = 0$ gives 1, while $x = 0$ gives -1
Lines are not enough	$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^4}$	every line gives 0, but $x = y^2$ gives $1/2$
Exists by squeezing	$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2+y^2} = 0$	absolute value is bounded above by $ y $

15 Practice problems

Concept checks

1. Explain why multivariable limits are harder than single-variable limits.
2. What does it mean for a function $f(x, y)$ to be continuous at (a, b) ?
3. Why can two different path limits prove that a limit does not exist?
4. Why does checking only horizontal and vertical paths usually not prove that a limit exists?
5. Explain the difference between a removable discontinuity and a path-dependent discontinuity.

Computational skill problems

6. Evaluate by substitution, if possible:

$$\lim_{(x,y) \rightarrow (2,-1)} (x^2y + e^{xy}).$$

7. Evaluate by substitution, if possible:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2}{x + y}.$$

8. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}.$$

9. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}.$$

10. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}.$$

11. Use polar coordinates to evaluate:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} + 1}.$$

12. Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0.$$

13. Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{\sqrt{x^2 + y^2}} = 0.$$

14. Determine where $f(x, y) = \frac{x + y}{x^2 + y^2 - 1}$ is continuous.

15. Determine where $g(x, y) = \sqrt{4 - x^2 - y^2}$ is continuous.

Modeling and interpretation problems

16. A temperature function $T(x, y)$ is continuous on a city map. Explain what continuity means physically.
17. A classifier is defined by $C(x, y) = 1$ if $x + y \geq 0$ and $C(x, y) = 0$ if $x + y < 0$. Where is this function discontinuous?
18. Let $L(w) = \|Aw - b\|^2$. Explain why L is continuous as a function of the parameter vector w .
19. Give an example of a model where discontinuity is expected or useful.
20. Give an example of a model where continuity is important for stability.

Computational practice

21. Test the paths $y = 2x$, $y = -2x$, and $y = 3x$ for $\frac{x^2 - y^2}{x^2 + y^2}$.
22. Test several paths for a new function of your choice.
23. Visualize the function $f(x, y) = \frac{xy}{x^2 + y^2}$ near the origin.
24. Numerically estimate the maximum of $\left| \frac{x^2 y}{x^2 + y^2} \right|$ on circles of radius r for several values of r .
25. Create a contour plot of a continuous loss function and explain why the level curves are useful.

16 Lab preview

The lab for this chapter focuses on three skills:

1. visualizing surfaces near singular points;
2. sampling limits along many paths;
3. distinguishing numerical evidence from mathematical proof.

A useful workflow is:

1. plot the surface or contour map;
2. test simple paths such as $y = 0$, $x = 0$, and $y = mx$;
3. test curved paths if needed;
4. use algebra to prove the final result.

Chapter summary

A multivariable limit requires agreement along all possible paths. Direct substitution works when a function is continuous at the point. To show that a limit does not exist, it is enough to find two paths with different limiting values. To show that a limit exists, we need a path-independent argument, often using estimates, polar coordinates, or the squeeze theorem.

Continuity means that the value of a function agrees with its local limiting behavior. This idea is central not only to calculus, but also to numerical modeling, optimization, statistics, data science, and machine learning.